

FORENSIC ANALYSIS REPORT

Case ID: CASE-0011

Analyst: John Doe

Generated On: 23/03/2026, 5:46:35 pm

System: ForenChain – Blockchain Chain of Custody

1. Evidence Details

Description: Dell XPS 15 laptop computer. Device was recovered from the suspect's residence (living room). Physical inspection confirmed the unique Service Tag: 5GHJ2 as labeled on the palm rest. The laptop has been secured on-site with tamper-evident forensic tape (Seal #77281) across all USB and power ports. Attached is official evidence tag #FC-2026-004B. Volatile RAM capture was completed prior to securing the device. Device is currently powered off and sealed.
Current Status: transferred_to_lab

2. Forensic Analysis

Tools Used:

- Autopsy (v4.20.0): Digital forensics platform used for deep file system analysis.Đ
- FTK Imager (v4.5.1): Tool used to acquire the bit-stream forensic image of the physical drive.Đ
- EnCase Endpoint Security: For memory forensics (RAM analysis).Đ
- HashCalc (v2.0): To verify SHA-256 data integrity hashes.Đ
- Tableau T35es (Write-Blocker): Physical hardware used to prevent data writing to the original drive.

Methodology:

1. Preparation: A clean, sterile forensic workstation was set up, and the Tableaus T35es physical write-blocker was initialized and verified.Đ
2. Image Acquisition: The hard drive was removed from the Dell XPS 15 laptop. FTK Imager was used to acquire an exact (bit-stream) forensic image of the drive into an E01 (EnCase) file format. A validation SHA-256 hash was generated at completion.Đ
3. Hash Verification: The hash of the newly acquired forensic image was verified against the MD5 hash generated during initial evidence logging. They matched, confirming integrity.Đ
4. Forensic Analysis: The forensic image was loaded into Autopsy and EnCase. The analysis focused on user activity logs, browser history, recently accessed files, and deleted data.

Observations:

- Browser Activity (Google Chrome): The suspect's browser history showed repeated visits to the "DarkNet" and forums discussing credit card skimming techniques on the date of 2026-03-20.Đ
- Deleted Files: Three encrypted containers (named backup1.vol, backup2.vol, and key_vault.log) were recovered from the \$Recycle.Bin that had been deleted on 2026-03-22 at 09:15 AM (approx. 30 minutes before the search warrant execution).Đ
- Registry Analysis: The Windows Registry analysis showed evidence that a USB storage device named "Lexar 32GB" was plugged in on 2026-03-21 at 23:30. This device was not found during the physical search.Đ
- Email Communication: Several outgoing emails with encrypted attachments were found, timestamped from March 18th to March 21st. The contents remain unreadable without decryption keys.

Conclusion:

Based on the forensic analysis of Case ID: CASE-0011, there is strong evidence to suggest the suspect was actively engaged in research and communication regarding digital financial fraud and credit card skimming. The recovery of deleted, encrypted containers from the workstation confirms attempts to conceal data prior to the search. The verification of the Lexar USB drive indicates a potential missing piece of evidence. The forensic integrity of the analysis was maintained at every step, with all actions and findings formally documented and anchored in the ForenChain ledger.

This forensic report is system-generated and digitally preserved. Any modification invalidates evidentiary

integrity.